

PRACTICAL - 05**Part A — Practice Programs (Hour 1)**

PROBLEM1	Shape areas: an abstract Shape with abstract area(); subclasses Circle, Rectangle, Triangle. Put a mix in a Shape[] and loop once to print each area and the running total, tracking the largest — one loop handling every shape through polymorphism.
CODE	<pre>abstract class Shape { public abstract double area(); } class Circle extends Shape { private double radius; public Circle(double radius) { this.radius = radius; } public double area() { return Math.PI * radius * radius; } } class Rectangle extends Shape { private double width, height; public Rectangle(double width, double height) { this.width = width; this.height = height; } public double area() { return width * height; } } class Triangle extends Shape { private double base, height;</pre>

```
public Triangle(double base, double height) {
    this.base = base;
    this.height = height;
}

public double area() {
    return 0.5 * base * height;
}

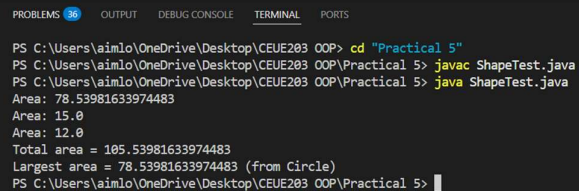
public class ShapeTest {

    public static void main(String[] args) {
        Shape[] shapes = {
            new Circle(5),
            new Rectangle(3, 5),
            new Triangle(4, 6),};
        double totalArea = 0;
        double largestArea = Double.MIN_VALUE;
        Shape largestShape = null;

        for (Shape s : shapes) {
            double area = s.area();
            System.out.println("Area: " + area);
            totalArea += area;

            if (area > largestArea) {
                largestArea = area;
                largestShape = s;
            }
        }

        System.out.println("Total area = " + totalArea);
        System.out.println("Largest area = " + largestArea + " (from " +
largestShape.getClass().getSimpleName() + ")");
    }
}
```

OUTPUT

```
PROBLEMS 36 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\aimlo\OneDrive\Desktop\CEUE203 OOP> cd "Practical 5"
PS C:\Users\aimlo\OneDrive\Desktop\CEUE203 OOP\Practical 5> javac ShapeTest.java
PS C:\Users\aimlo\OneDrive\Desktop\CEUE203 OOP\Practical 5> java ShapeTest.java
Area: 78.53981633974483
Area: 15.0
Area: 12.0
Total area = 105.53981633974483
Largest area = 78.53981633974483 (from Circle)
PS C:\Users\aimlo\OneDrive\Desktop\CEUE203 OOP\Practical 5> |
```

PROBLEM2	Payroll: an abstract Employee with abstract monthlySalary(); FullTime (fixed), PartTime (hours × rate) and Intern (stipend), each calling super for the shared name/id. Loop a mixed array to print each salary and the total; use instanceof to add a note only for interns.
CODE	<pre>abstract class Employee { private String name; private int id; public Employee(String name, int id) { this.name = name; this.id = id; } public String getName() { return name; } public int getId() { return id; } public abstract double monthlySalary(); } class FullTime extends Employee { private double salary; public FullTime(String name, int id, double salary) { super(name, id); this.salary = salary; } public double monthlySalary() { return salary; } } class PartTime extends Employee { private int hours; private double rate;</pre>

```
public PartTime(String name, int id, int hours, double rate) {
    super(name, id);
    this.hours = hours;
    this.rate = rate;
}

public double monthlySalary() {
    return hours * rate;
}
}

class Intern extends Employee {

    private double stipend;

    public Intern(String name, int id, double stipend) {
        super(name, id);
        this.stipend = stipend;
    }

    public double monthlySalary() {
        return stipend;
    }
}

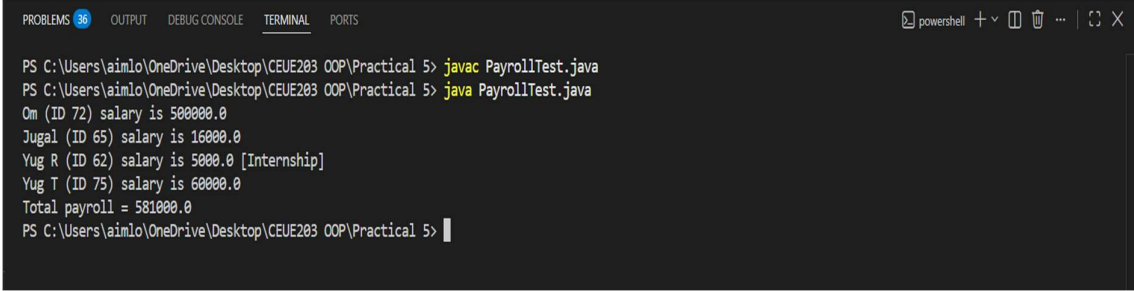
public class PayrollTest {

    public static void main(String[] args) {
        Employee[] employees = {
            new FullTime("Om", 72, 500000),
            new PartTime("Jugal", 65, 80, 200),
            new Intern("Yug R", 62, 5000),
            new FullTime("Yug T", 75, 60000),};

        double total = 0;

        for (Employee e : employees) {
            double salary = e.monthlySalary();
            System.out.print(e.getName() + " (ID " + e.getId() + ") salary is " + salary);
            if (e instanceof Intern) {
                System.out.print(" [Internship]");
            }
            System.out.println();
            total += salary;
        }

        System.out.println("Total payroll = " + total);
    }
}
```

	<pre> } } </pre>
OUTPUT	 <pre> PS C:\Users\aimlo\OneDrive\Desktop\CEUE203 OOP\Practical 5> javac PayrollTest.java PS C:\Users\aimlo\OneDrive\Desktop\CEUE203 OOP\Practical 5> java PayrollTest.java Om (ID 72) salary is 500000.0 Jugal (ID 65) salary is 16000.0 Yug R (ID 62) salary is 5000.0 [Internship] Yug T (ID 75) salary is 60000.0 Total payroll = 581000.0 PS C:\Users\aimlo\OneDrive\Desktop\CEUE203 OOP\Practical 5> </pre>
PROBLEM3	(For Advanced Learners – home) Model a few media types that each compute a late fee differently and total the fees for a returned batch. Design the hierarchy and rules yourself.
CODE	<pre> abstract class Media { private String title; private int id; public Media(String title, int id) { this.title = title; this.id = id; } public String getTitle() { return title; } public int getId() { return id; } public abstract double lateFee(int daysLate); } class Book extends Media { public Book(String title, int id) { super(title, id); } @Override public double lateFee(int daysLate) { return daysLate * 2.0; } } class DVD extends Media { public DVD(String title, int id) { super(title, id); } } </pre>

```
public double lateFee(int daysLate) {
    return daysLate * 5.0;
}

class Magazine extends Media {
    public Magazine(String title, int id) {
        super(title, id);
    }

    public double lateFee(int daysLate) {

        return daysLate > 0 ? 10.0 : 0.0;
    }
}

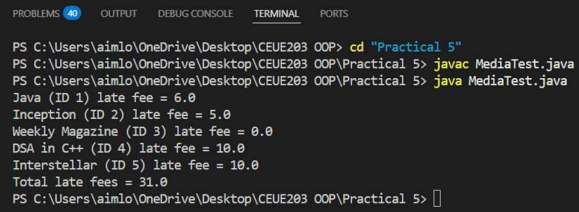
public class MediaTest {
    public static void main(String[] args) {
        Media[] items = {
            new Book("Java", 1),
            new DVD("Inception", 2),
            new Magazine("Weekly Magazine", 3),
            new Book("DSA in C++", 4),
            new DVD("Interstellar", 5)
        };

        int[] daysLate = {3, 1, 0, 5, 2};
        double totalFee = 0;

        for (int i = 0; i < items.length; i++) {
            double fee = items[i].lateFee(daysLate[i]);
            System.out.println(items[i].getTitle() + " (ID " + items[i].getId() +
                ") late fee = " + fee);
            totalFee += fee;
        }

        System.out.println("Total late fees = " + totalFee);
    }
}
```

OUTPUT



```
PS C:\Users\aimlo\OneDrive\Desktop\CEUE203 OOP> cd "Practical 5"
PS C:\Users\aimlo\OneDrive\Desktop\CEUE203 OOP\Practical 5> javac MediaTest.java
PS C:\Users\aimlo\OneDrive\Desktop\CEUE203 OOP\Practical 5> java MediaTest.java
Java (ID 1) late fee = 6.0
Inception (ID 2) late fee = 5.0
Weekly Magazine (ID 3) late fee = 0.0
DSA in C++ (ID 4) late fee = 10.0
Interstellar (ID 5) late fee = 10.0
Total late fees = 31.0
PS C:\Users\aimlo\OneDrive\Desktop\CEUE203 OOP\Practical 5> 
```